

Spherical Harmonics, Symmetry, Entropy, and Fractal Structure in non-Born–Oppenheimer Chemistry and Quantum Space-Time with Thought Experiments Extending to Consciousness

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Abstract

Spherical harmonics arise as eigenfunctions of angular momentum operators associated with the rotation group $SO(3)$ and organize the angular structure of quantum states in three-dimensional space. Their usual role as basis functions in separated-coordinate treatments captures only part of their significance. When rotational symmetry is considered together with permutation symmetry and fully correlated non-Born–Oppenheimer formulations, the admissible Hilbert space becomes a constrained subset of the naive tensor-product construction.

The familiar degeneracy $2\ell + 1$ defines angular microstate multiplicity and therefore an entropy scale, but that count is modified once coupled angular momentum, exchange symmetry, and all-particle correlation are imposed simultaneously. The resulting state space is recursively generated, symmetry-pruned, and asymptotically sparse relative to the unrestricted representation space. Such behavior admits a fractal interpretation in the sense of hierarchical construction, non-uniform occupancy, and sublinear effective scaling dimension.

This structure extends beyond representation theory. The symmetry-constrained Hilbert space may be interpreted as an information-theoretic structure with reduced entropy, as a symmetry-defined encoded subspace within a larger quantum system, and as a restriction on physically admissible measurement outcomes. More generally, these results imply that physically realizable quantum states occupy a structured and symmetry-filtered subset of the ambient Hilbert space.

This perspective aligns with recent approaches that model cognition and consciousness using mathematically constrained dynamical systems, in which coherence, stability, and transition behavior emerge from underlying structural restrictions. Within the present framework, such phenomena may be reinterpreted at the level of state space: coherence corresponds not to unrestricted superposition, but to dynamics confined to symmetry-compatible subspaces with reduced entropy and constrained accessibility.

The discussion develops this framework at the level of representation theory, state counting, non-Born–Oppenheimer molecular physics, and constrained information geometry, with an explicit illustration based on the non-Born–Oppenheimer hydrogen molecule.

Keywords— Spherical harmonics, Non-Born–Oppenheimer methods, Permutation symmetry, Angular momentum coupling, Hilbert space structure, Quantum entropy, Fractal scaling, Many-body quantum systems



Motivation

The standard formulation of quantum mechanics provides a complete and internally consistent description of physical systems, yet many of its structural consequences remain under-interpreted. In particular, the organization of Hilbert space under symmetry—especially rotational symmetry, permutation symmetry, and many-body correlation—is typically treated as a technical device for simplifying calculations rather than as a fundamental constraint on the space of physically realizable states.

In conventional treatments, angular momentum theory introduces spherical harmonics as basis functions and establishes degeneracy factors such as $2\ell + 1$ as bookkeeping results associated with irreducible representations of $SO(3)$. Similarly, tensor-product constructions and coupling schemes are used to build many-body states, while permutation symmetry is imposed as an additional constraint to enforce particle statistics. These elements are well understood individually, but they are rarely considered together as defining a global structure on Hilbert space itself.

This raises a natural question: to what extent does the physically admissible Hilbert space differ from the naive tensor-product construction suggested by unconstrained degrees of freedom?

The motivation for this work is to address this question by examining how rotational symmetry, angular momentum coupling, permutation symmetry, and fully correlated non-Born–Oppenheimer structure interact to define a constrained and structured subset of Hilbert space. When these ingredients are considered simultaneously, the admissible state space is not merely reduced in size, but reorganized in a way that is recursive, non-uniform, and asymptotically sparse relative to the ambient construction space.

This perspective suggests that familiar quantities—such as degeneracy, entropy, and state count—may be reinterpreted not simply as properties of individual sectors, but as measures of how symmetry filters and organizes the global state space. In particular, the reduction in admissible states under combined symmetry constraints motivates the introduction of scaling concepts and the possibility of an effective dimensionality smaller than that of the naive tensor-product space.

The goal of the present work is therefore to make this structure explicit. By combining representation-theoretic analysis, non-Born–Oppenheimer quantum chemistry, and information-theoretic interpretation, a framework is developed in which the physically admissible Hilbert space is understood as a symmetry-constrained, recursively generated object with nontrivial scaling behavior.

This reinterpretation is not intended to replace standard quantum mechanics, but to reorganize its well-established components into a unified structural picture. In doing so, it provides a basis for connecting symmetry, entropy, state-space geometry, and information, and for exploring how these constraints manifest across domains ranging from molecular physics to structured dynamical systems such as processes that might be governing consciousness.



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1 Introduction

Rotational symmetry is one of the most persistent organizing principles in quantum mechanics. In three-dimensional space, the existence of a distinguished action of the rotation group on state space ensures that angular momentum is not merely an observable among many others, but part of the kinematic architecture of the theory. Once this structure is imposed, spherical harmonics emerge unavoidably as the angular eigenfunctions associated with irreducible representations of $SO(3)$ [1, 2, 3]. In standard pedagogy, their significance is usually introduced through separation of variables in central potentials. That route is mathematically correct, but it can obscure the deeper fact that spherical harmonics define a family of symmetry sectors, each with a specific degeneracy, transformation law, and information content.

The angular momentum labels (ℓ, m) are usually interpreted as quantum numbers attached to single-particle orbitals or to total angular states obtained after coupling. However, those labels also define admissible channels of state organization. For fixed ℓ , the multiplicity $2\ell + 1$ is often treated as a bookkeeping result, but it may be read equally well as the number of distinguishable angular microstates available within that irreducible representation. That interpretation introduces entropy and information capacity directly into the angular sector of Hilbert space. In that sense, angular momentum is not only geometric but informational.

The situation becomes substantially richer once the analysis moves beyond single-particle or Born–Oppenheimer treatments. In a fully non-Born–Oppenheimer framework, all particles are quantum mechanical from the outset. Electrons and nuclei are placed on the same formal footing, and the wavefunction is defined on the full configuration space rather than on a hierarchy of conditional subspaces. Under these conditions, spherical harmonics no longer serve simply as independent factors attached to decoupled coordinates. Instead, angular structure becomes distributed across correlated degrees of freedom. Once permutation symmetry is imposed on identical particles, entire sectors of the naive tensor-product Hilbert space are removed. The admissible space is then determined by the simultaneous action of rotational symmetry, exchange symmetry, and all-particle correlation.

Such a constrained space has several noteworthy features. First, its accessible state count is smaller than the naive count suggested by unconstrained angular degeneracies. Second, its construction is recursive: the addition of each degree of freedom enlarges the tensor-product space, but subsequent symmetry projection prunes it. Third, the fraction of admissible states can decrease with scale. Those three properties together motivate a fractal interpretation. The term “fractal” is used here in a structural and asymptotic sense: a recursively generated set with non-uniform occupancy and effective scaling dimension smaller than the ambient construction space. This behavior emerges from recursive construction followed by symmetry-induced pruning.

The goal of the following discussion is therefore not to replace standard angular momentum theory, but to reinterpret and extend it into a broader structural framework. The first task is to establish the representation-theoretic role of spherical harmonics and the entropy associated with angular degeneracy. The second is to show how those ideas are modified by non-Born–Oppenheimer structure and generalized permutation symmetry. The third is to develop a scaling framework in which the constrained Hilbert space can be analyzed as a recursively pruned object with fractal-like features.



These developments are then extended through complementary perspectives involving information theory, quantum encoding, measurement, and combinatorial structure. An explicit discussion of the hydrogen molecule without the Born–Oppenheimer approximation is used to anchor the abstract framework in a concrete quantum-chemical system.

The central claim of this work is that the physically admissible Hilbert space of many-body quantum systems is a recursively generated, symmetry-constrained subset of the naive tensor-product construction, exhibiting sublinear scaling and fractal-like structure.

The significance of this structure is not limited to representation theory or quantum chemistry alone. The same constraints that reduce admissible state count also admit complementary interpretations across multiple domains. In particular, the symmetry-constrained Hilbert space may be understood in information-theoretic terms as a reduced-entropy configuration space, in quantum-informational terms as an encoded subspace defined by intrinsic constraints, in combinatorial terms as a recursively pruned branching structure analogous to binomial expansion, and in measurement theory as a restriction on physically realizable outcomes. These perspectives do not introduce new assumptions, but instead reflect different projections of the same underlying structure.

Recent work has proposed that consciousness may be described by mathematically constrained dynamical systems exhibiting critical behavior and structured phase transitions [4]. The present work complements this perspective by showing that, independent of dynamical assumptions, the underlying Hilbert space itself is constrained by symmetry and many-body structure, thereby restricting the set of physically realizable states to a reduced and structured subset of the ambient space.

2 Angular Momentum and Representation Structure

Rotational symmetry in three-dimensional space is generated by angular momentum operators satisfying the commutation relations:

$$[L_i, L_j] = i\hbar\epsilon_{ijk}L_k, \quad (1)$$

where L_i ($i = x, y, z$) denote the components of the angular momentum operator, $\hbar$ is the reduced Planck constant, and ϵ_{ijk} is the Levi–Civita symbol encoding the antisymmetric structure of the rotation algebra [2, 5].

These commutators define the Lie algebra $\mathfrak{so}(3)$ and therefore determine how state vectors transform under infinitesimal rotations [6, 7]. The associated quadratic Casimir operator is:

$$L^2 = L_x^2 + L_y^2 + L_z^2, \quad (2)$$

and because L^2 commutes with all three generators, it classifies irreducible representations.

To make the structure explicit, consider spherical coordinates (r, θ, ϕ) , where $r \in [0, \infty)$ is the radial coordinate, $\theta \in [0, \pi]$ is the polar angle measured from the positive z -axis, and $\phi \in [0, 2\pi)$ is the



azimuthal angle in the xy -plane.

The angular dependence of quantum states is described by spherical harmonics $Y_\ell^m(\theta, \phi)$, which arise as simultaneous eigenfunctions of the squared angular momentum operator L^2 and its z -component L_z .

Here, ℓ denotes the total angular momentum quantum number, taking non-negative integer values $\ell \in \mathbb{N}_0$, and determines the irreducible representation of $\text{SO}(3)$. For each fixed ℓ , the integer m labels the magnetic quantum number, corresponding to the projection of angular momentum along the z -axis, with $m \in \{-\ell, -\ell + 1, \dots, \ell\}$.

The pair (ℓ, m) therefore uniquely specifies a basis element within the $(2\ell + 1)$ -dimensional irreducible representation space associated with angular momentum ℓ . The simultaneous eigenfunctions of L^2 and L_z are the spherical harmonics $Y_\ell^m(\theta, \phi)$, satisfying [3]:

$$L^2 Y_\ell^m = \ell(\ell + 1) \hbar^2 Y_\ell^m, \quad (3)$$

$$L_z Y_\ell^m = m \hbar Y_\ell^m, \quad (4)$$

with

$$\ell \in \mathbb{N}_0, \quad m \in \{-\ell, -\ell + 1, \dots, \ell\}. \quad (5)$$

For each ℓ , the representation space has dimension:

$$g_\ell = 2\ell + 1. \quad (6)$$

Here g_ℓ denotes the degeneracy of the angular momentum sector labeled by ℓ , that is, the number of distinct and mutually orthogonal quantum states corresponding to different values of the magnetic quantum number m within the same irreducible representation [8]. This structure is illustrated schematically in Figure 1, where each angular momentum sector appears as a set of $2\ell + 1$ distinguishable states labeled by m .

This degeneracy arises directly from the structure of the $\text{SO}(3)$ representation: for fixed ℓ , the allowed values of m range from $-\ell$ to ℓ in integer steps, yielding $2\ell + 1$ independent states.

From a Hilbert-space perspective, g_ℓ is therefore the dimension of the irreducible subspace $\mathcal{H}_\ell$ associated with angular momentum ℓ . That dimension is not incidental. It is dictated by the structure of the irreducible representation itself.

At the level of Hilbert-space organization, rotational invariance therefore induces a decomposition [9]. Let $\mathcal{H}$ denote the Hilbert space of square-integrable complex-valued functions defined on the unit sphere in three-dimensional space. The sphere is denoted by

$$S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\},$$

where $\mathbb{R}^3$ denotes three-dimensional real Euclidean space (the set of all triples of real numbers), and S^2 denotes the two-dimensional unit sphere (the superscript referring to dimension, not to spin momentum or entropy in following sections). This space represents the set of admissible angular wavefunctions in three-dimensional quantum systems.

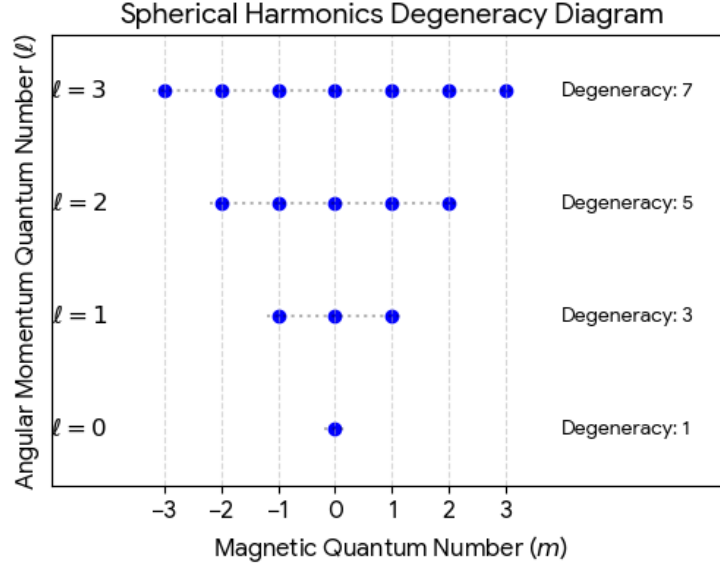


Figure 1: Angular momentum sectors for $\ell = 0, 1, 2, 3$, showing the degeneracy $g_\ell = 2\ell + 1$ as the number of distinct magnetic quantum numbers m . Each horizontal level corresponds to an irreducible representation H_ℓ of $SO(3)$.

Under the action of the rotation group, $\mathcal{H}$ decomposes into a direct sum of invariant subspaces associated with irreducible representations of $SO(3)$ [5, 6]:

$$\mathcal{H} = \bigoplus_{\ell=0}^{\infty} \mathcal{H}_\ell, \quad \dim(\mathcal{H}_\ell) = 2\ell + 1. \quad (7)$$

Each subspace $\mathcal{H}_\ell$ is invariant under rotations, and the total space may be regarded as an ordered stack of angular sectors indexed by ℓ . Such a decomposition is familiar in central-potential problems, but its structural role remains important even when variables do not separate neatly. The angular labels still define symmetry sectors; what changes in more complicated systems is how those sectors are embedded in the full many-body space.

The usual derivation of spherical harmonics begins from the angular part of the Laplacian in spherical coordinates. That derivation is pedagogically useful, but the representation-theoretic perspective clarifies something deeper: spherical harmonics are not merely solutions to a differential equation, but basis elements for a symmetry-determined decomposition of state space. This is important because entropy, state counting, and symmetry projection all operate on the basis of accessible sectors rather than on a purely differential notion of angular dependence.

Orthogonality also matters here. The spherical harmonics satisfy [8]:

$$\int_{S^2} Y_\ell^m(\theta, \phi)^* Y_{\ell'}^{m'}(\theta, \phi) d\Omega = \delta_{\ell\ell'} \delta_{mm'}. \quad (8)$$

Here δ_{ij} denotes the Kronecker delta, defined by $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ otherwise. This orthogonality means that the (ℓ, m) labels correspond to distinguishable directions in Hilbert



space. Once information-theoretic language is introduced, orthogonality becomes a statement of distinguishability and therefore an ingredient in state counting.

This decomposition establishes the angular sectors that define the initial combinatorial structure of the Hilbert space. The next step is to interpret this multiplicity in informational terms. These symmetry-defined angular sectors provide the combinatorial structure on which entropy and information measures will be defined in Section 3: Angular Degeneracy, Entropy, and Information Capacity.

3 Angular Degeneracy, Entropy, and Information Capacity

The degeneracy associated with a fixed angular momentum sector naturally defines a count of distinguishable configurations. Because the spherical harmonics Y_ℓ^m form an orthonormal basis of $\mathcal{H}_\ell$, each value of m corresponds to a distinct and distinguishable quantum state. This motivates the definition of the angular microstate count. For fixed ℓ , the degeneracy $g_\ell = 2\ell + 1$ gives the number of distinct magnetic sublevels available in the absence of additional symmetry breaking. Interpreting this multiplicity as a count of distinguishable angular microstates:

$$\Omega_\ell = 2\ell + 1, \quad (9)$$

where Ω_ℓ denotes the number of distinguishable angular microstates within the irreducible representation labeled by ℓ , and the corresponding entropy (S_ℓ) is the natural logarithm ($\ln$) of the degeneracy (Ω_ℓ) [10, 11]:

$$S_\ell = k_B \ln(2\ell + 1). \quad (10)$$

Here k_B denotes Boltzmann's constant, which sets the scale for entropy when expressed in physical units. This quantity may be understood as the entropy associated with the angular sector alone. It does not require thermal equilibrium in the usual statistical-mechanical sense; rather, it arises from the combinatorial multiplicity of accessible orthogonal angular states.

It is important to distinguish this entropy from total thermodynamic entropy. The expression above does not account for radial excitations, spin multiplicities, vibrational structure, or environmental couplings. Instead, it isolates the entropy contribution associated with angular degeneracy. That isolation is useful because it exposes a direct bridge between symmetry and information. From an information-theoretic perspective, the degeneracy also determines the number of bits required to specify a state within a fixed angular momentum sector. Because there are $\Omega_\ell = 2\ell + 1$ distinguishable states, the corresponding information content is given by [10]:

$$I_\ell = \log_2(2\ell + 1), \quad (11)$$

where I_ℓ denotes the information required to uniquely identify a state within the ℓ sector, measured in bits. From this point of view, each ℓ sector carries a definite angular information capacity [10, 12]. This relationship reflects the general principle illustrated in Figure 2, where lower-probability configurations carry higher information content. In this setting, degeneracy may be interpreted



as defining a uniform distribution over admissible states, for which Shannon entropy reduces to a logarithmic state-counting expression.

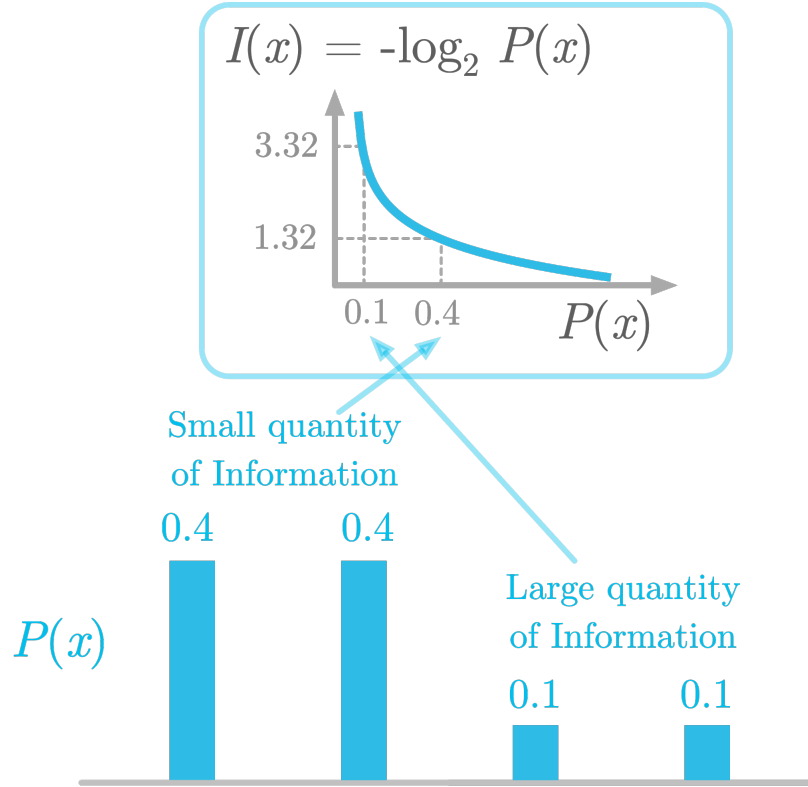


Figure 2: Here $P(x)$ denotes the probability of a particular configuration x , and $I(x)$ the associated Shannon information. Relationship between probability $P(x)$ and information content $I(x) = -\log_2 P(x)$. Rare events carry higher information, while more probable events carry less information.

That observation becomes meaningful when asking how much of the nominal angular state space remains available once further constraints are imposed. If all m values are physically admissible and dynamically reachable, then the entropy S_ℓ describes the full angular contribution. If additional constraints remove some subset of states, then the effective angular entropy must be replaced by:

$$S_{\ell,\text{eff}} = k_B \ln \tilde{g}_\ell, \quad \tilde{g}_\ell \leq 2\ell + 1. \quad (12)$$

The problem therefore becomes one of constrained state counting. The naive degeneracy $2\ell + 1$ is only the starting point.

This way of framing the problem also shifts the interpretation of symmetry. Rotational symmetry does not merely label states. It determines the coarse geometry of the state space. Additional symmetries then act as filters on that geometry. Entropy is therefore not an independent quantity layered on top of the Hilbert space; it is a measure of how much of that geometry remains accessible under the full set of constraints.

Once entropy is associated with angular degeneracy, the key question becomes how this entropy is modified under additional constraints such as coupling and permutation symmetry. The question



then becomes how this angular entropy is modified once full many-body structure and symmetry constraints are introduced, as discussed in Section 5: Non-Born–Oppenheimer Structure and Loss of Separability.

4 Time Evolution and Symmetry-Preserving Dynamics

The significance of angular sectors becomes clearer when dynamics are introduced. For a rotationally invariant Hamiltonian (H):

$$[H, L^2] = 0, \quad [H, L_z] = 0, \quad (13)$$

the angular momentum quantum numbers are conserved. Time evolution is generated by the unitary operator $U(t)$, which satisfies $U(t)^\dagger U(t) = I$, where $U(t)^\dagger$ denotes the adjoint (Hermitian conjugate) and I is the identity operator [5]:

$$U(t) = e^{-iHt/\hbar}, \quad (14)$$

and because H commutes with L^2 , each angular sector evolves independently:

$$U(t)\mathcal{H}_\ell \subseteq \mathcal{H}_\ell. \quad (15)$$

This has an immediate informational meaning. A system initialized in one ℓ sector cannot dynamically migrate to another unless rotational symmetry is broken. The symmetry therefore partitions the state space not only statically but dynamically. The space of accessible trajectories is smaller than the full Hilbert space. Invariant subspaces become dynamically isolated channels.

The point becomes more pronounced in many-body systems. There the full Hilbert space may be enormous, but symmetry still carves it into sectors that do not communicate under symmetric evolution. If additional symmetries are present, such as exchange symmetry, the accessible trajectories contract further. In that sense, dynamics do not merely occur in Hilbert space; they occur in a symmetry-filtered subspace of Hilbert space.

Selection rules furnish another illustration. Transition matrix elements of the form:

$$\langle \psi_f | \hat{O} | \psi_i \rangle \quad (16)$$

vanish unless the operator O transforms compatibly with the representations carried by the initial and final states [1, 5, 8]. The Wigner–Eckart structure underlying such statements shows that symmetry limits not only the existence of states but also the existence of transitions. Information flow is therefore restricted at both the kinematic and dynamical levels.

Thus, symmetry not only partitions the state space but restricts its dynamical accessibility. This restriction becomes more pronounced when the full many-body structure is considered.



5 Non-Born–Oppenheimer Structure and Loss of Separability

The starting point for a fully correlated description of atomic and molecular systems is the many-body Hamiltonian, written in Laboratory Cartesian coordinate frame (LCCF) as:

$$H = \sum_{i=1}^N -\frac{\hbar^2}{2M_i} \nabla_{\mathbf{R}_i}^2 + \sum_{i<j} V_{ij} \|\mathbf{R}_i - \mathbf{R}_j\| \quad (17)$$

where N is the total number of particles (for all nuclei and electrons), $\mathbf{R}_i \in \mathbb{R}^3$ denotes the position of the i -th particle and M_i its mass [2]. The corresponding wavefunction $\Psi(\mathbf{R}_1, \dots, \mathbf{R}_N)$ is defined over the full configuration space in LCCF and contains all spatial correlations between particles.

In the Born–Oppenheimer approximation, the wavefunction is factorized as:

$$\Psi(\mathbf{R}_{\text{elec}}, \mathbf{R}_{\text{nuc}}) \approx \psi_{\text{elec}}(\mathbf{R}_{\text{elec}}; \mathbf{R}_{\text{nuc}}) \chi_{\text{nuc}}(\mathbf{R}_{\text{nuc}}) \quad (18)$$

where $\mathbf{R}_{\text{elec}} = \{\mathbf{R}_{\text{elec}_1}, \dots, \mathbf{R}_{\text{elec}_{\text{total}}}\}$ denotes the collection of electronic coordinates and $\mathbf{R}_{\text{nuc}} = \{\mathbf{R}_{\text{nuc}_1}, \dots, \mathbf{R}_{\text{nuc}_{\text{total}}}\}$ the collection of nuclear coordinates, with each $\mathbf{R}_{\text{elec}_i}, \mathbf{R}_{\text{nuc}_j} \in \mathbb{R}^3$ [13]. This approximation reduces the full many-body problem to a sequence of parametrized electronic problems. However, such a separation neglects explicit correlations between electronic and nuclear motion.

The ground state of the system is obtained through the variational principle, which states that the expectation value of the Hamiltonian:

$$E = \frac{\langle \Psi | H | \Psi \rangle}{\langle \Psi | \Psi \rangle} \quad (19)$$

is minimized by the exact ground-state wavefunction [5].

Laboratory and Internal Coordinate Representations

To isolate the physically relevant structure, it is standard to introduce the center-of-mass coordinate:

$$\mathbf{R}_{\text{CM}} = \frac{1}{M} \sum_{i=1}^N M_i \mathbf{R}_i \quad (20)$$

where $M = \sum_{i=1}^N M_i$ is the total mass of the system (electrons and nuclei included) [14, 15]. In free space, the motion of the center of mass separates from the internal degrees of freedom and does not affect the spectral properties of the system. As a result, observable energy levels and transition frequencies are determined entirely by the internal Hamiltonian.

The coordinate set $\{\mathbf{r}\}_{i=1}^N - 1$, with $\mathbf{r}_i \in \mathbb{R}^3$, defines the system in the Internal Cartesian coordinate frame (CCF) and when this shift happens, the heaviest particle becomes the center of the internal CCF system (and is labeled particle "0"). While this formulation is convenient for writing the Hamiltonian, it does not contain redundant degrees of freedom associated with global translation.



The wavefunction is expressed directly as:

$$\Psi = \Psi(\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_{N-1}) \quad (21)$$

with all correlations encoded in its functional form.

This transformation is illustrated schematically in Figure 3, where the laboratory coordinate frame is separated into center-of-mass and internal degrees of freedom.

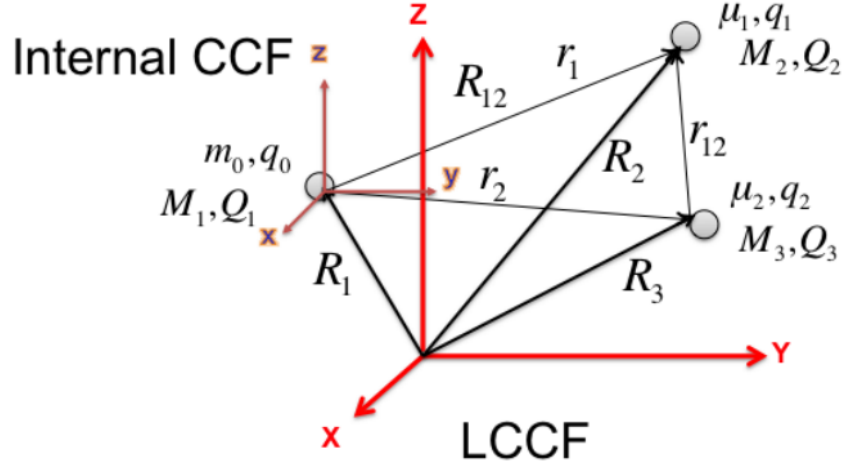


Figure 3: Transformation from laboratory-frame coordinates (LCCF) to internal coordinates (CCF), illustrating the separation of center-of-mass motion and the definition of translationally invariant internal coordinates. Adapted from Sharkey [16, 17, 18].

In Figure 3, the symbols M_i and Q_i denote the masses and charges of the constituent particles in the LCCF, and in the internal CCF system these quantities become μ_i (the reduced mass) and $q_i = Q_i$ charge. The quantities $\mathbf{r}_i$ and $\mathbf{R}_i$ represent position vectors in the internal and laboratory coordinate frame (respectively).

The remaining degrees of freedom are described by translationally invariant internal coordinates. One may introduce a set of $(3N - 3)$ independent internal coordinates $\{\mathbf{r}_a\}_{a=1}^{N-1}$ constructed from linear combinations of particle positions that are independent of $\mathbf{R}_{\text{CM}}$.

After separation of the center-of-mass motion, the wavefunction may be written in the form

$$\Psi(\mathbf{R}_1, \dots, \mathbf{R}_N) = \Psi_{\text{CM}}(\mathbf{R}_{\text{CM}}) \Psi_{\text{int}}(\mathbf{r}_1, \dots, \mathbf{r}_{N-1}) \quad (22)$$

where Ψ_{CM} describes translational motion of the system as a whole and Ψ_{int} encodes all physically relevant structure.

In the internal-coordinate representation, the total orbital angular momentum operator acts on the translationally invariant degrees of freedom rather than on the center-of-mass coordinate. After separation of the center-of-mass motion, the physically relevant angular momentum operator is



therefore

$$\mathbf{L}_{\text{int}} = \sum_{i=1}^{N-1} \mathbf{r}_i \times (-i\hbar \nabla_{\mathbf{r}_i}) \quad (23)$$

where $\{\mathbf{r}_i\}$ denotes any independent set of internal coordinates and $\nabla_{\mathbf{r}_i}$ is the gradient with respect to $\mathbf{r}_i$ [19].

The operators L_{int}^2 and $L_{\text{int},z}$ generate rotations of the internal configuration. It is their eigenstructure—not that of the laboratory-frame center-of-mass motion—that organizes the admissible angular sectors of the correlated many-body wavefunction. In this sense, spherical-harmonic structure persists in the non-Born–Oppenheimer setting, but it is no longer attached to independent single-particle coordinates. Instead, it acts on the internal configuration space of the system as a whole. In the non-Born–Oppenheimer setting, angular momentum is therefore a property of the internal many-body configuration rather than of independently separable single-particle coordinates.

Accordingly, permutation symmetry and angular momentum coupling act on the same internal many-body configuration space, which is why state construction and state counting must be formulated after transformation to internal coordinates. All angular momentum operators and symmetry constraints act on the internal component Ψ_{int} , not on the center-of-mass degree of freedom. Consequently, the analysis of angular structure, permutation symmetry, and correlation is naturally formulated in the internal coordinate representation.

This perspective clarifies that the physically admissible Hilbert space is defined not in the full laboratory coordinate space, but in a reduced internal configuration space on which symmetry and correlation act intrinsically.

Once the full configuration space is retained, angular dependence is no longer globally separable as

$$\psi(r, \theta, \phi) = R(r)Y_{\ell}^m(\theta, \phi) \quad (24)$$

for each subsystem independently. Angular information becomes distributed across correlated degrees of freedom.

A common example is the explicitly correlated Gaussian (ECG) expansion [20, 21, 22]:

$$\Psi(\mathbf{r}) = \sum_k c_k \exp\left(-\mathbf{r}^T A_k \mathbf{r}\right), \quad (25)$$

where $\mathbf{r} = (r_1, \dots, r_{N-1}) \in \mathbb{R}^{3(N-1)}$, and A_k are positive-definite matrices encoding interparticle correlation with the dimension of $3(N-1) \times 3(N-1)$. Such basis functions do not present spherical harmonics explicitly as separate factors, yet angular momentum remains present and must be imposed through the global action of the angular momentum operator. The angular structure is therefore not absent; it is encoded non-locally across the internal configuration space.

This loss of separability is crucial. Once angular degrees of freedom are no longer attached cleanly to independent particle coordinates, state counting cannot proceed by simply multiplying single-particle degeneracies. Coupling and exchange constraints become intrinsic to the angular description itself.

The loss of separability implies that angular structure can no longer be treated independently



for each particle, necessitating a fully coupled and symmetry-aware description. This observation leads directly to the role of permutation symmetry in defining the admissible state space, which is developed in Section 6: Permutation Symmetry and Generalized Pauli Constraints.

This breakdown of separability is precisely what invalidates naive multiplicative state counting based on independent angular sectors, necessitating a coupled and symmetry-constrained description of the full many-body state space.

The separation of center-of-mass motion is not merely a technical simplification, but a structural step that identifies the physically relevant domain of the many-body problem. In free space, the center-of-mass coordinate contributes only a trivial translational degree of freedom, leaving the internal coordinates to determine all spectroscopic and dynamical properties. Consequently, the physically admissible Hilbert space is not defined on the full laboratory-coordinate space, but on the reduced internal configuration space.

This distinction is essential for the arguments developed in the present work. The symmetry constraints discussed in subsequent sections—rotational symmetry, permutation symmetry, and angular momentum coupling—act on the internal degrees of freedom, not on the center-of-mass motion. As a result, the recursive construction and symmetry-induced pruning of state space occur entirely within the internal Hilbert space. The reduction in admissible state count and the emergence of nontrivial scaling behavior therefore arise from structure internal to the system, rather than from trivial kinematic degrees of freedom.

In this sense, the separation of center-of-mass motion isolates the space in which symmetry acts non-trivially, providing the foundation for the constrained Hilbert-space framework developed throughout the remainder of the paper.

6 Permutation Symmetry and Generalized Pauli Constraints

The permutation symmetry discussed in this section acts on the same internal configuration space introduced in Section 5. After separation of the center-of-mass motion, all physically relevant degrees of freedom reside in the internal Hilbert space, and it is on this space that both rotational symmetry and permutation symmetry act simultaneously. Consequently, the admissible state space must be constructed as a joint symmetry-compatible subspace of the internal many-body configuration space.

For any set of identical particles, the wavefunction must transform according to the permutation group S_N [6, 23]. If $\hat{P}_\pi$ denotes the operator associated with permutation π , then physical states satisfy:

$$\hat{P}_\pi \Psi = \pm \Psi, \quad (26)$$

with the sign determined by particle statistics. The associated projectors are the symmetrizer:

$$\mathcal{S} = \frac{1}{N!} \sum_{\pi \in S_N} \hat{P}_\pi \quad (27)$$



and the antisymmetrizer:

$$\mathcal{A} = \frac{1}{N!} \sum_{\pi \in S_N} \text{sgn}(\pi) \hat{P}_\pi. \quad (28)$$

These operators project the naive Hilbert space onto subspaces compatible with bosonic or fermionic exchange symmetry [9, 23].

This projection does not merely discard isolated basis vectors. It removes entire representation sectors of the permutation group. As a result, the physical space is:

$$\mathcal{H}_{\text{phys}} = \mathcal{A}\mathcal{H} \quad \text{or} \quad \mathcal{S}\mathcal{H}, \quad (29)$$

and therefore:

$$\dim(\mathcal{H}_{\text{phys}}) < \dim(\mathcal{H}) \quad (30)$$

whenever nontrivial exchange constraints are present.

The phrase “generalized Pauli constraints” is appropriate here because the exclusion mechanism is not limited to simple orbital occupancy statements [24]. In correlated many-body spaces, anti-symmetry acts globally. Once angular momentum coupling is included, admissibility is determined by compatibility between rotational symmetry and exchange symmetry. The relevant structure is therefore not just $SO(3)$ or just S_N , but their combined action. The physical subspace must lie inside the joint symmetry-compatible sector of:

$$SO(3) \times S_N. \quad (31)$$

This joint action already suggests why naive degeneracy counts become unreliable. The factor $2\ell + 1$ describes the representation dimension associated with $SO(3)$ alone. Exchange symmetry can reduce the number of physically admissible states even within a fixed angular momentum sector. What remains is a smaller state set with reduced entropy and altered scaling.

This separation clarifies that the physically admissible Hilbert space is defined not in the full laboratory coordinate space, but in a reduced internal configuration space on which symmetry and correlation act intrinsically. The effect of these symmetry constraints becomes particularly clear when combined with angular momentum coupling, which introduces a recursive structure discussed in Section 7: Coupled Angular Momentum and Recursive Many-Body Construction.

7 Coupled Angular Momentum and Recursive Many-Body Construction

Once multiple angular momenta are present, the relevant Hilbert space is not a single $\mathcal{H}_\ell$ but a tensor product of such spaces [8, 25]. For two subsystems,

$$\mathcal{H}_{\ell_1} \otimes \mathcal{H}_{\ell_2} = \bigoplus_{L=|\ell_1-\ell_2|}^{\ell_1+\ell_2} \mathcal{H}_L. \quad (32)$$



At the level of basis functions, the transformation is described by Clebsch–Gordan coefficients [25]:

$$Y_{\ell_1}^{m_1} \otimes Y_{\ell_2}^{m_2} = \sum_{L,M} C_{\ell_1 m_1 \ell_2 m_2}^{LM} Y_L^M. \quad (33)$$

This process generalizes recursively. For three subsystems,

$$(\mathcal{H}_{\ell_1} \otimes \mathcal{H}_{\ell_2}) \otimes \mathcal{H}_{\ell_3} = \left(\bigoplus_{L_{12}} \mathcal{H}_{L_{12}} \right) \otimes \mathcal{H}_{\ell_3} = \bigoplus_{L_{12}} \bigoplus_L \mathcal{H}_L. \quad (34)$$

This recursive coupling structure defines the combinatorial expansion of angular momentum sectors prior to the imposition of symmetry constraints. It is this expansion—followed by symmetry-induced pruning—that underlies the effective reduction in admissible state count discussed in subsequent sections.

This recursive coupling structure may be visualized through analogous examples from electron spin addition ($s = 1/2$), as illustrated in Figure 4. In such diagrams, the total spin (S) evolves through successive addition of individual spin degrees of freedom, producing a branching structure in which each node represents a possible total spin sector.

At each step, the addition of a new degree of freedom enlarges the set of admissible states, while the coupling rules constrain the allowed combinations, resulting in a structured branching pattern. Although the figure is expressed in terms of spin, the same recursive structure applies to orbital angular momentum coupling, where irreducible representation sectors are generated through tensor-product composition and subsequent projection.

The key feature of this construction is that the number of admissible pathways grows combinatorially, but not all branches remain physically valid once symmetry constraints are imposed. This observation provides a concrete visualization of the recursive expansion that precedes symmetry-induced pruning, which becomes central to the scaling arguments developed in later sections.

Permutation symmetry then acts on this branching structure. Some branches survive; others are eliminated. The result is naturally represented as a tree in which nodes correspond to intermediate angular momentum sectors and pruning corresponds to exchange-incompatible combinations. The state-space structure is therefore recursive by construction.

This recursive point of view is central for the fractal interpretation. Fractal behavior does not emerge from angular momentum alone, nor from exchange symmetry alone, but from repeated enlargement by tensor product followed by repeated reduction by symmetry projection. The qualitative mechanism is the same at each step:

- (i) enlarge the space by adding degrees of freedom,
- (ii) recouple angular sectors,
- (iii) prune by symmetry.

What changes with system size is the combinatorial complexity of the branching. This recursive

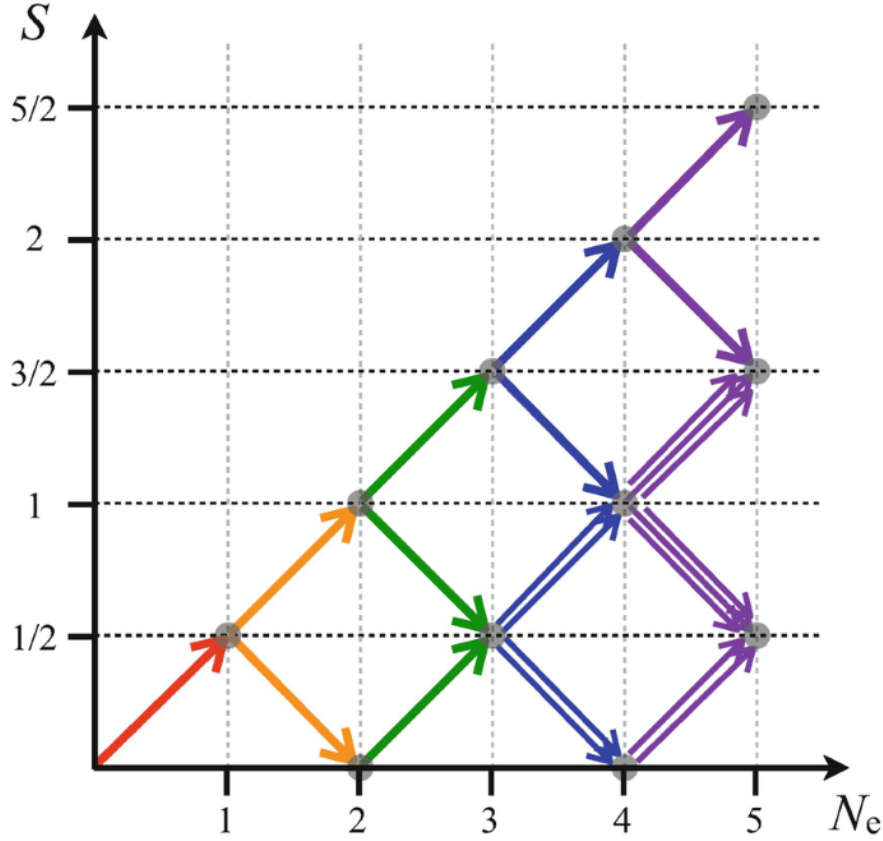


Figure 4: Recursive coupling of spin degrees of freedom illustrating the branching structure generated by successive addition of angular momentum-like quantities. Each node represents a total spin sector S obtained through coupling, and the branching pattern reflects the combinatorial structure of tensor-product composition. While shown here for spin, the same recursive structure applies to orbital angular momentum coupling in many-body systems.

construction provides the combinatorial foundation for the scaling analysis developed in Section 9 and 11.

8 Representation-Theoretic Form of the Physical Hilbert Space

The full many-body Hilbert space may be written as

$$\mathcal{H}_{\text{full}} = \bigotimes_{i=1}^N \mathcal{H}^{(i)}, \quad (35)$$

with each factor carrying some rotational representation. Decomposing each factor into irreducible sectors gives

$$\mathcal{H}^{(i)} = \bigoplus_{\ell_i} \mathcal{H}_{\ell_i}, \quad (36)$$



and therefore

$$\mathcal{H}_{\text{full}} = \bigoplus_{\{\ell_i\}} \left(\bigotimes_{i=1}^N \mathcal{H}_{\ell_i} \right). \quad (37)$$

Each tensor product then decomposes into total-angular-momentum sectors:

$$\bigotimes_{i=1}^N \mathcal{H}_{\ell_i} = \bigoplus_L \mathcal{H}_L \otimes \mathcal{M}_{\{\ell_i\}}^{(L)}, \quad (38)$$

where $\mathcal{M}_{\{\ell_i\}}^{(L)}$ is a multiplicity space accounting for the different ways of obtaining total angular momentum L .

Permutation symmetry acts nontrivially on these multiplicity spaces. The physically admissible space is therefore of the form

$$\mathcal{H}_{\text{phys}} = \bigoplus_L \mathcal{H}_L \otimes \left(\mathcal{M}_{\{\ell_i\}}^{(L)} \cap \mathcal{H}_{\text{sym}} \right), \quad (39)$$

where $\mathcal{H}_{\text{sym}}$ denotes the subspace with the required exchange symmetry. Here $\mathcal{M}_{\{\ell_i\}}^{(L)}$ denotes the multiplicity space associated with total angular momentum L , accounting for the number of distinct coupling pathways through which L may be obtained from the set $\{\ell_i\}$. This expression makes the reduction mechanism explicit. Rotational symmetry determines the available total-angular-momentum sectors. Permutation symmetry filters the multiplicity spaces. The result is not just a smaller state count, but a smaller family of representation sectors.

That form is useful because it clarifies where the entropy reduction occurs. The reduction is not solely in the $2L + 1$ factor associated with the irreducible $SO(3)$ representation. It also occurs in the multiplicity spaces, which can be substantially reduced or even eliminated by exchange constraints. In other words, the admissible state count is shaped both by the dimension of each irreducible representation of $SO(3)$ and the multiplicities permitted by symmetry constraints.

9 Explicit Non-Born–Oppenheimer Example: The Hydrogen Molecule

An explicit example helps anchor the discussion. Consider the hydrogen molecule treated without the Born–Oppenheimer approximation [20]. The system contains two electrons and two protons. The full wavefunction depends on all four particle coordinates [21]:

$$\Psi = \Psi(\mathbf{r}_1, \mathbf{r}_2, \mathbf{R}_1, \mathbf{R}_2). \quad (40)$$

The Hamiltonian is

$$H = -\frac{\hbar^2}{2m_e}(\nabla_1^2 + \nabla_2^2) - \frac{\hbar^2}{2m_p}(\nabla_{R_1}^2 + \nabla_{R_2}^2) + V, \quad (41)$$



where

$$V = \frac{e^2}{|\mathbf{r}_1 - \mathbf{r}_2|} + \frac{e^2}{|\mathbf{R}_1 - \mathbf{R}_2|} - \sum_{i=1}^2 \sum_{j=1}^2 \frac{e^2}{|\mathbf{r}_i - \mathbf{R}_j|}. \quad (42)$$

Rotational symmetry implies the existence of a total angular momentum operator

$$\mathbf{L}_{\text{tot}} = \mathbf{L}_e + \mathbf{L}_p, \quad (43)$$

where $\mathbf{L}_e$ refers to the electronic contribution and $\mathbf{L}_p$ to the nuclear contribution in a suitable translationally invariant coordinate system. In a coupled description, admissible states are organized by the total quantum numbers (L, M) .

However, rotational symmetry is not the only constraint. The two electrons are identical fermions, so the electronic part of the total wavefunction must be antisymmetric under exchange. The two protons are also identical particles, and their allowed exchange symmetry depends on nuclear spin coupling. The physical space therefore satisfies

$$\mathcal{H}_{\text{phys}} \subset \mathcal{H}_{SO(3)} \cap \mathcal{H}_{S_2^{(e)}} \cap \mathcal{H}_{S_2^{(p)}}. \quad (44)$$

A naive angular count would proceed as though four independent angular sectors contributed multiplicatively, suggesting scaling of the form

$$\Omega_{\text{naive}}(\ell) \sim (2\ell + 1)^4. \quad (45)$$

That estimate is too large. In the actual non-Born–Oppenheimer problem, not every combination of electronic and nuclear angular momenta is admissible. Coupling rules eliminate some combinations; antisymmetry eliminates others; proton exchange symmetry can remove additional sectors depending on the spin state. The result is an effective state count

$$N_{H_2}(\ell) \ll (2\ell + 1)^4. \quad (46)$$

The physical meaning is clear. The non-Born–Oppenheimer hydrogen molecule does not merely occupy a four-factor product of unconstrained angular spaces. Its angular structure is a correlated and symmetry-filtered subset of that larger construction. This is precisely the type of reduction needed to motivate a nontrivial scaling analysis.

10 State Counting Under Simultaneous Constraints

The contrast between naive and physical state counts may be formalized as follows. For N degrees of freedom, if each contributes a sector of size roughly $2\ell + 1$, then the unconstrained count is

$$\Omega_{\text{naive}}(\ell, N) \sim (2\ell + 1)^N. \quad (47)$$



Once total angular momentum coupling and exchange symmetry are imposed, the admissible count becomes

$$N(\ell, N) \ll (2\ell + 1)^N. \quad (48)$$

It is convenient to define the occupancy ratio

$$f(\ell, N) = \frac{N(\ell, N)}{(2\ell + 1)^N}. \quad (49)$$

If there were no additional constraints, then $f(\ell, N) = 1$. In the constrained problem, one expects

$$0 < f(\ell, N) < 1. \quad (50)$$

The key asymptotic question is whether

$$f(\ell, N) \rightarrow 0 \quad \text{as} \quad \ell \rightarrow \infty \quad (51)$$

for fixed or moderately growing N . If so, then the admissible state set occupies a vanishing fraction of the naive state space at large angular scale.

Such behavior is plausible because two reduction mechanisms intensify with scale. First, the number of coupling pathways grows rapidly with the number of factors, making multiplicity spaces large but structurally constrained. Second, exchange symmetry acts globally on those multiplicity spaces, and the fraction compatible with the required symmetry need not remain constant. In effect, the number of ways to build a state increases, but the number of ways to build an admissible state can grow more slowly.

Hydrogen Atom Energy Level Convergence

The structure of the hydrogen spectrum provides a concrete illustration of how dynamical laws and symmetry constraints jointly organize the admissible state space. In hydrogenic systems, the bound-state energies follow the well-known scaling

$$E_n = -\frac{R_H}{n^2}, \quad (52)$$

where n is the principal quantum number and R_H is the Rydberg constant. As $n \rightarrow \infty$, the energy levels approach the ionization threshold,

$$\lim_{n \rightarrow \infty} E_n = 0^-, \quad (53)$$

at which point the discrete bound-state spectrum merges into the continuum.

The spacing between adjacent levels correspondingly decreases as

$$\Delta E_n = E_{n+1} - E_n \sim \frac{1}{n^3}, \quad (54)$$



so that the spectrum becomes increasingly dense near the ionization limit [26]. This accumulation reflects a general property of quantum systems in which discrete eigenvalues converge toward the onset of a continuous spectrum. At the same time, the degeneracy of hydrogenic states grows with increasing principal quantum number,

$$g_n \sim n^2, \quad (55)$$

indicating that the number of admissible states increases even as their energy spacing decreases.

This behavior is illustrated in Figure 5, where the difference between each bound-state energy and the ionization limit decreases systematically with increasing n . The plot shows that the spectrum does not fill the energy axis uniformly, but instead exhibits a structured accumulation governed by the underlying Coulomb dynamics.

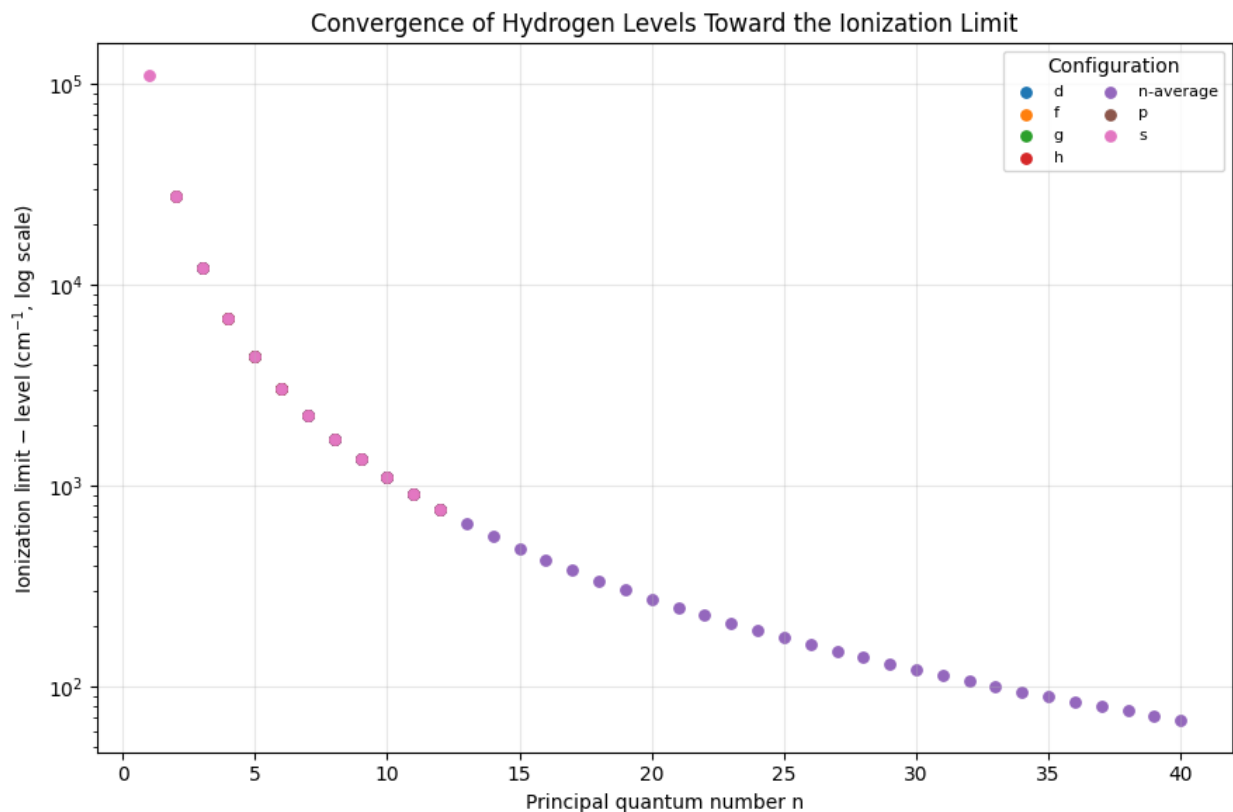


Figure 5: Convergence of hydrogen energy levels toward the ionization limit using NIST ASD data [27]. The decreasing separation from the ionization threshold reflects the hydrogenic $E_n \sim -1/n^2$ scaling, while the density of states increases with principal quantum number n .

Within the present framework, this convergence may be interpreted as a spectral manifestation of constrained state organization. While the energy scaling itself arises from the Coulomb potential, the admissible states at each level are not arbitrary. They are structured by angular momentum coupling, permutation symmetry, and many-body constraints, which restrict the accessible Hilbert space to a symmetry-compatible subset of the full tensor-product construction.

In this sense, the hydrogen spectrum illustrates the coexistence of dynamical and structural effects.



The Coulomb interaction determines the asymptotic behavior of the energy levels, while symmetry determines how states are organized, labeled, and constrained within each energy manifold. The accumulation of states near the ionization limit therefore occurs within a structured and symmetry-filtered Hilbert space, rather than within an unconstrained combinatorial expansion.

This example provides a concrete bridge between standard spectral theory and the constrained Hilbert-space framework developed in this work. It shows that even in the simplest quantum system, the physically realizable states form an ordered and restricted subset of the ambient space, motivating the analysis of how such constraints affect state counting and scaling behavior in more general many-body systems.

11 Fractal Scaling and Effective Dimension

This projection introduces a global constraint that acts across the entire Hilbert space, fundamentally altering the combinatorial structure established earlier. To make the scaling statement precise, define the effective dimension

$$D_{\text{eff}}(\ell, N) = \frac{\log N(\ell, N)}{\log(2\ell + 1)^N}. \quad (56)$$

If the limit exists, then

$$D_{\text{eff}}(N) = \lim_{\ell \rightarrow \infty} \frac{\log N(\ell, N)}{\log(2\ell + 1)^N}. \quad (57)$$

In the unconstrained case, $N(\ell, N) \sim (2\ell + 1)^N$ and therefore $D_{\text{eff}} = 1$ relative to the ambient tensor-product scaling. In the constrained case, suppose

$$N(\ell, N) \sim (2\ell + 1)^{\alpha N}, \quad 0 < \alpha < 1. \quad (58)$$

Then

$$D_{\text{eff}} = \alpha. \quad (59)$$

Sublinear scaling in the exponent thus corresponds directly to an effective dimension smaller than that of the ambient construction. These structural features motivate a fractal interpretation in the sense of recursively generated, non-uniformly occupied sets with reduced effective dimension [28].

This pruning structure is illustrated schematically in Figure 6, where the unconstrained combinatorial expansion is contrasted with the symmetry-filtered subset of admissible states.

The language of fractality is appropriate here for three reasons. First, the admissible set is generated recursively: tensor product, recoupling, symmetry projection. Second, the admissible set can occupy a vanishing fraction of the larger construction space as scale increases. Third, the resulting effective dimension can be non-integer relative to the naive scaling dimension. These are precisely the structural features that motivate fractal descriptions in many other mathematical settings.

It is not necessary to claim literal geometric self-similarity in coordinate space. The claim is instead that the admissible state space in representation-theoretic coordinates is recursively organized and asymptotically sparse in a way that is naturally described by fractal scaling. The ambient space is

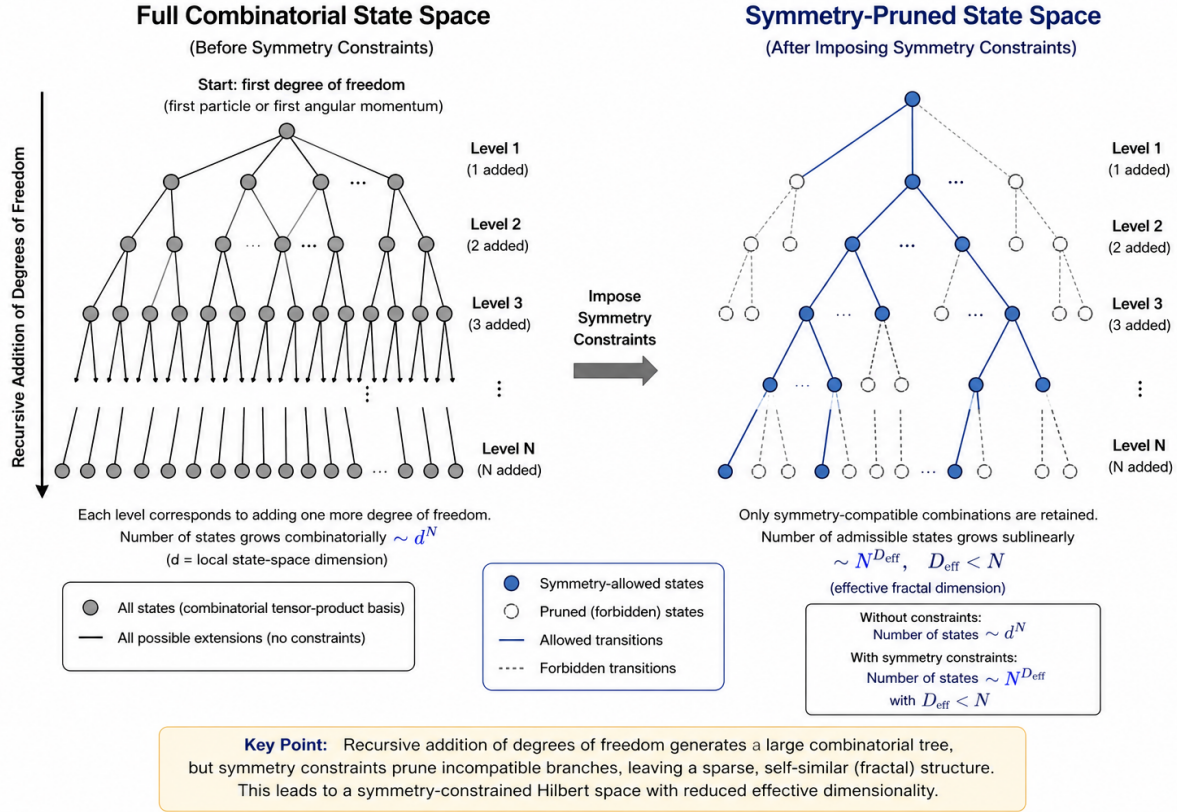


Figure 6: Recursive branching of the Hilbert space under tensor-product construction (left) and the corresponding symmetry-pruned structure (right). While the unconstrained space grows combinatorially, symmetry constraints remove incompatible branches, leaving a structured and asymptotically sparse subset. This recursive pruning underlies the effective reduction in accessible state count and motivates a fractal interpretation of the constrained Hilbert space.

Hilbert-space construction space, not Euclidean embedding space.

The fractal characterization is therefore not imposed, but emerges directly from the repeated interaction between combinatorial expansion and symmetry constraint.

12 Recursive Pruning as the Mechanism Behind Fractal Behavior

The mechanism behind the scaling can be made more explicit through recursion. Suppose a constrained N -body space $\mathcal{H}_N^{\text{phys}}$ has already been constructed. The addition of one more degree of freedom first enlarges the space by tensor product:

$$\mathcal{H}_{N+1}^{\text{naive}} = \mathcal{H}_N^{\text{phys}} \otimes \mathcal{H}_1. \quad (60)$$



That operation multiplies the local state count roughly by a factor of $2\ell + 1$. The next step recouples angular momentum sectors:

$$\mathcal{H}_{N+1}^{\text{naive}} \rightarrow \bigoplus_L \mathcal{H}_L \otimes \mathcal{M}^{(L)}. \quad (61)$$

The final step projects onto the required exchange-symmetry sector:

$$\mathcal{H}_{N+1}^{\text{phys}} = \mathcal{P}_{\text{sym}} \left(\bigoplus_L \mathcal{H}_L \otimes \mathcal{M}^{(L)} \right). \quad (62)$$

The key point is that the projection acts after branching has occurred. The state space is therefore not built by smoothly filling an ambient continuum. It is built by repeated branching followed by selective deletion. That is the natural habitat of fractal-like sets. The same pattern repeats at each level of construction, but with increasing combinatorial complexity.

If the average retention factor after projection is denoted by $r(\ell, N)$, then one may write schematically

$$N(\ell, N+1) \sim r(\ell, N) (2\ell + 1) N(\ell, N), \quad (63)$$

with

$$0 < r(\ell, N) < 1. \quad (64)$$

If $r(\ell, N)$ remains bounded away from 1 in an appropriate asymptotic sense, then repeated multiplication by a retention factor smaller than unity drives the effective exponent below the naive value. The exact asymptotic law depends on the detailed representation content, but the qualitative mechanism is robust.

13 Spectral Consequences of Constrained Hilbert Spaces

A constrained Hilbert space has spectral consequences because the Hamiltonian acts only within the admissible subspace:

$$H : \mathcal{H}_{\text{phys}} \rightarrow \mathcal{H}_{\text{phys}}. \quad (65)$$

Energy eigenstates satisfy

$$H\Psi_n = E_n\Psi_n, \quad (66)$$

but the available eigenvectors are fewer than they would be in the unconstrained space. This has at least three implications.

First, the spectral density is reduced. If $N(E)$ denotes the number of eigenstates below energy E , then one expects

$$N(E)_{\text{phys}} < N(E)_{\text{full}} \quad (67)$$

whenever the unconstrained space is formally larger.

Second, degeneracy patterns are altered. Rotational symmetry alone yields the familiar factor $2L + 1$, but exchange symmetry can reduce the number of admissible states within a given total- L



sector. Thus the effective degeneracy satisfies

$$\tilde{g}_L \leq 2L + 1. \quad (68)$$

The spectrum therefore reflects not only the rotational symmetry of the Hamiltonian but also the exchange structure of the physical state space.

Third, transition structure is modified. Selection rules already constrain transitions under symmetry-preserving operators, but once admissible sectors are fewer, some transitions that would exist formally in the unconstrained space do not correspond to any physical state pair at all. The symmetry-pruned Hilbert space therefore induces a symmetry-pruned transition graph.

14 Information Geometry of the Constrained State Space

The constrained state space may also be understood geometrically. Let

$$\mathcal{S}_{\text{phys}} = \{\Psi \in \mathcal{H}_{\text{phys}} : \|\Psi\| = 1\} \quad (69)$$

denote the unit sphere inside the admissible Hilbert space. Distances between pure states can be measured by the Fubini–Study metric, which may be written in its simplest form as

$$d(\Psi_1, \Psi_2) = \arccos |\langle \Psi_1 | \Psi_2 \rangle|. \quad (70)$$

Because $\mathcal{S}_{\text{phys}}$ lies inside a symmetry-filtered subspace, the physically accessible geometry is not the geometry of the full unit sphere in the ambient Hilbert space. It is the geometry of an embedded subset defined by simultaneous algebraic constraints.

This perspective helps interpret entropy reduction. Reduced state count means not only fewer basis vectors but a smaller accessible information manifold. If the admissible states are recursively distributed inside the ambient space with vanishing asymptotic occupancy, then the accessible geometry is sparse and highly structured. The fractal language therefore has an informational meaning as well as a combinatorial one.

A complementary information-theoretic formulation of this structure is developed in Appendix A: Information-Theoretic Structure of Constrained Hilbert Spaces.

15 Implications for Constrained-State Encoding

The same mathematical features that reduce the physical state count also shape the space of admissible encodings. If only states satisfying rotational symmetry, exchange symmetry, and coupled-angular-momentum constraints are physically allowed, then arbitrary superpositions drawn from the ambient unconstrained construction space are not physically meaningful within the system. Allowed encodings must live in

$$\mathcal{H}_{\text{phys}} \subsetneq \mathcal{H}_{\text{full}}. \quad (71)$$



This immediately implies a reduction in accessible configuration count. If information is stored in state labels drawn from the admissible set, then the raw capacity is

$$I(\ell, N) = \log_2 N(\ell, N), \quad (72)$$

not

$$I_{\max}(\ell, N) = \log_2(2\ell + 1)^N. \quad (73)$$

The difference

$$\Delta I(\ell, N) = I_{\max}(\ell, N) - I(\ell, N) \quad (74)$$

is an information deficit induced by symmetry.

Such a deficit is not merely a limitation. It also expresses structural robustness. Because not every formally expressible state is admissible, arbitrary perturbations need not map one admissible state to another admissible state. The system's state graph is constrained. Any interpretation involving verification, constrained transitions, or structured encoding must therefore be formulated in the smaller physical space. The key point is that the restriction is intrinsic. It arises from symmetry and many-body structure rather than from an external rule layered onto the system.

16 Structural Synthesis and Emergent Scaling Behavior

The discussion above develops a chain of implications that begins with spherical harmonics and ends with a symmetry-pruned, recursively generated, fractal-like Hilbert space. The logical sequence is now clear.

First, spherical harmonics define the irreducible angular sectors of $SO(3)$, each with degeneracy $2\ell + 1$. Second, that degeneracy defines an angular entropy and information capacity. Third, rotationally invariant dynamics preserve those sectors, so the angular decomposition has dynamical significance. Fourth, a fully non-Born–Oppenheimer formulation distributes angular structure across all particles and removes the convenience of independent single-particle angular labels. Fifth, permutation symmetry projects out exchange-incompatible sectors, reducing multiplicities and therefore reducing accessible state count. Sixth, coupled angular momentum introduces a recursive branching structure that is repeatedly pruned by symmetry at each stage of many-body construction. Seventh, the resulting state count can scale sublinearly relative to the naive ambient tensor-product count, yielding an effective dimension smaller than the ambient one.

Nothing in this sequence depends on authorial preference or interpretive imposition. The structure follows from the interaction of rotational symmetry, exchange symmetry, and full many-body quantum mechanics. The fractal language does not replace standard quantum mechanics; it provides a compact description of the recursively generated and asymptotically sparse structure that emerges from those standard ingredients.

These experimentally observable constraints demonstrate that physically realizable quantum states occupy a structured and reduced subset of the ambient Hilbert space, providing an operational



foundation for the broader interpretive framework developed below.

The structural framework developed above provides a natural point of contact with existing proposals that attribute a role to quantum processes in cognition and consciousness. In particular, models proposed by Stuart Hameroff and Roger Penrose [29] suggest that conscious experience arises from coordinated quantum processes distributed across neuronal structures, as further elaborated in subsequent work [30]. Consequently, any quantum-mechanical account of consciousness, if such a description is valid, would be expected to operate within such a constrained manifold.

Such approaches, including recent formulations based on mathematically constrained dynamical systems [4], typically invoke large-scale coherence and global integration but do not explicitly specify the structure of the underlying Hilbert space in which these processes occur. The present analysis suggests that, in any many-body quantum system, the physically admissible state space is not the full tensor-product construction, but a recursively generated, symmetry-constrained subspace characterized by reduced entropy and sublinear scaling.

Within this framework, the notion of global coherence may be reinterpreted as structured occupancy of a symmetry-compatible subspace of the full Hilbert space. The loss of separability in fully correlated non-Born–Oppenheimer systems implies that admissible states are intrinsically non-factorizable, aligning with the requirement of integrated behavior across degrees of freedom. However, this integration does not occur within an unconstrained space; it is restricted to a subset defined by the joint action of rotational symmetry, permutation symmetry, and many-body correlation.

Within this framework, the notion of global coherence may be reinterpreted as structured occupancy of a symmetry-compatible subspace of the full Hilbert space. The loss of separability in fully correlated non-Born–Oppenheimer systems implies that admissible states are intrinsically non-factorizable, aligning with the requirement of integrated behavior across degrees of freedom. However, this integration does not occur within an unconstrained space; it is restricted to a subset defined by the joint action of rotational symmetry, permutation symmetry, and many-body correlation.

If consciousness is modeled as a dynamical system embedded in a quantum-mechanical substrate, then its admissible states would be constrained by the same symmetry and many-body structure that govern physical Hilbert spaces. In this sense, the present framework provides structural conditions under which such models must be formulated.

Consequently, any quantum-mechanical account of consciousness would be expected to operate within such a constrained manifold.

Consequently, any quantum-mechanical account of consciousness must operate within such a constrained manifold. The effective reduction in accessible state count and the associated entropy deficit imply that admissible configurations form a structured and asymptotically sparse subset of the ambient space. From this perspective, quantum models of consciousness may be understood not as invoking arbitrary coherence, but as implicitly requiring dynamics confined to a symmetry-filtered Hilbert space whose structure is determined by fundamental principles of quantum mechanics.



Experimental Manifestation in Molecular Hydrogen The symmetry-constrained Hilbert-space framework admits direct experimental interpretation in molecular spectroscopy. In the non-Born–Oppenheimer description, H_2 is a four-particle quantum system whose admissible states must satisfy rotational symmetry, electron-exchange symmetry, proton-exchange symmetry, and all-particle correlation simultaneously. As a result, the physically accessible state space is a strict subset of the naive tensor-product construction.

A naive angular construction would suggest a state count scaling as $(2\ell + 1)^4$. However, symmetry constraints eliminate entire sectors of this space. In particular, proton-exchange symmetry couples nuclear-spin symmetry to rotational parity: para- H_2 occupies even rotational levels, while ortho- H_2 occupies odd rotational levels. Thus, entire rotational sectors are absent for a given spin isomer, providing a direct manifestation of symmetry-induced pruning.

This reduction is observable in high-resolution rovibrational spectra. The experimentally measured spectrum contains not only allowed transitions, but also systematic absences, altered degeneracy patterns, and a restricted transition graph relative to the unconstrained case. These features reflect the underlying symmetry-filtered structure of the admissible Hilbert space.

A practical validation pathway is therefore to construct the symmetry-admissible set of states labeled by (v, J, I, S) , compare the resulting transition graph against empirical rovibrational data, and benchmark transition energies against non-Born–Oppenheimer explicitly correlated Gaussian calculations. Agreement across these levels would provide an operational test of the symmetry-pruned Hilbert-space framework.

17 Conclusion

Spherical harmonics are most often introduced as angular basis functions, but their role is deeper than basis convenience. They define symmetry sectors whose degeneracy determines angular microstate multiplicity and associated entropy. When the analysis is extended to fully correlated non-Born–Oppenheimer systems and constrained further by permutation symmetry, the physically admissible Hilbert space becomes a proper subset of the naive tensor-product construction. Coupled angular momentum creates a recursive branching in representation space, and symmetry projection prunes that branching repeatedly. The resulting space is hierarchically organized, asymptotically sparse, and naturally described by an effective scaling dimension smaller than that of the ambient construction space. In that sense, the admissible state space exhibits fractal-like structure arising directly from symmetry in three-dimensional quantum systems.

The appendices provide complementary perspectives on the constrained Hilbert-space structure developed in the main text, including its information-theoretic formulation, its interpretation as an encoded subspace, and its relation to combinatorial and representation-theoretic constructions.

The preceding discussion has focused on the structural properties of constrained Hilbert spaces arising from symmetry and many-body quantum mechanics. The appendices that follow develop complementary perspectives on this structure, including its interpretation in information-theoretic terms, its relation to encoded subspaces in quantum computation, its implications for measurement,



and its underlying combinatorial and representation-theoretic foundations.

A Information-Theoretic Structure of Constrained Hilbert Spaces

Appendix A extends the entropy-based discussion of Section 3 by introducing an explicit Shannon information formulation.

The preceding discussion develops entropy primarily from state counting, where multiplicity defines an effective measure of accessible configurations. In this appendix, that perspective is extended by introducing an explicit information-theoretic formulation based on Shannon entropy. This provides a complementary description of constrained Hilbert spaces in terms of probability distributions rather than purely combinatorial counts.

A.1 Probability Measures on Admissible State Spaces

Let $\mathcal{H}_{\text{phys}}$ denote the physically admissible Hilbert space obtained after imposing rotational symmetry, permutation symmetry, and full non-Born–Oppenheimer correlation. Consider a discrete orthonormal basis $\{|\psi_i\rangle\}$ spanning this space, with cardinality

$$N(\ell, N) = \dim(\mathcal{H}_{\text{phys}}). \quad (75)$$

A probability distribution over admissible states is defined by a set $\{p_i\}$ satisfying

$$p_i \geq 0, \quad \sum_{i=1}^{N(\ell, N)} p_i = 1. \quad (76)$$

The Shannon entropy associated with this distribution is

$$H = - \sum_{i=1}^{N(\ell, N)} p_i \log p_i. \quad (77)$$

This quantity measures the information required to specify the state of the system within the constrained Hilbert space, given the probability distribution $\{p_i\}$.

A.2 Relation to State Counting

In the special case of a uniform distribution,

$$p_i = \frac{1}{N(\ell, N)}, \quad (78)$$

the Shannon entropy reduces to

$$H = \log N(\ell, N). \quad (79)$$



This recovers the logarithmic state-counting expression used in the main text. However, the general formulation allows for non-uniform distributions, which arise naturally when symmetry, dynamics, or interaction structure favor certain regions of Hilbert space.

If the unconstrained Hilbert space were accessible, the corresponding maximal entropy would be

$$H_{\max} = \log(2\ell + 1)^N. \quad (80)$$

Under constraints,

$$H \leq \log N(\ell, N) < \log(2\ell + 1)^N, \quad (81)$$

with strict inequality whenever symmetry removes admissible states or induces non-uniform weighting.

A.3 Entropy Reduction Under Symmetry Constraints

The reduction of entropy relative to the unconstrained case may be written as

$$\Delta H = \log(2\ell + 1)^N - H. \quad (82)$$

This quantity measures the information deficit induced by symmetry constraints. It reflects two distinct mechanisms:

1. **State elimination:** entire sectors of the Hilbert space are removed by permutation symmetry and angular momentum coupling rules.
2. **Probability redistribution:** even within the admissible set, symmetry and dynamics can produce non-uniform occupation probabilities.

Both effects contribute to reducing the effective entropy relative to the naive tensor-product estimate.

A.4 Connection to Fractal Scaling

The constrained state count $N(\ell, N)$ exhibits sublinear scaling relative to the naive estimate $(2\ell + 1)^N$. This motivates the definition of an effective dimension:

$$D_{\text{eff}} = \lim_{\ell \rightarrow \infty} \frac{\log N(\ell, N)}{\log(2\ell + 1)^N}. \quad (83)$$

In the uniform-distribution limit, Shannon entropy satisfies

$$H \sim D_{\text{eff}} \log(2\ell + 1)^N. \quad (84)$$

Thus, the Shannon entropy directly probes the effective dimensionality of the admissible Hilbert



space. When $D_{\text{eff}} < 1$, the entropy grows sublinearly relative to the ambient tensor-product space, reflecting the asymptotic sparsity of admissible states.

For non-uniform distributions, the entropy is further reduced:

$$H \leq D_{\text{eff}} \log(2\ell + 1)^N, \quad (85)$$

indicating that both structural constraints and probability concentration contribute to the effective reduction of accessible information.

A.5 Relation to Holographic Information Bounds (Interpretive Note)

The symmetry-constrained reduction of the admissible Hilbert space discussed in this work suggests a broader class of information-theoretic structures in which the effective number of degrees of freedom is smaller than that of the ambient construction space. In a different context, related ideas appear in holographic formulations of quantum gravity, where the information content of a region is conjectured to scale with its boundary rather than its volume [31, 32, 33].

The present framework does not assume or derive holographic duality. Rather, the comparison is structural: both settings exhibit a reduction of effective degrees of freedom and a restriction on admissible state configurations relative to a naive counting argument.

A.6 Interpretation

The introduction of Shannon entropy clarifies that the constrained Hilbert space is not only smaller in cardinality but also structured in terms of information distribution. The recursive construction and symmetry-induced pruning discussed in the main text imply that admissible states are not uniformly distributed within the ambient space. Instead, they form a hierarchically organized subset whose informational content scales with an effective dimension smaller than that of the full tensor-product construction.

This interpretation reinforces the fractal characterization developed earlier. The constrained Hilbert space may be regarded as an information-bearing structure in which both the number of states and their distribution reflect underlying symmetry principles. Shannon entropy provides a natural quantitative measure of this structure without requiring additional physical assumptions beyond those already imposed by symmetry and quantum mechanics.

A.7 Scope and Limitations

The entropy discussed here is informational rather than thermodynamic. No assumption of thermal equilibrium is required, and the probabilities $\{p_i\}$ need not correspond to a canonical or microcanonical ensemble. The formulation applies equally to pure-state ensembles, dynamical distributions, or abstract probability assignments over admissible states.

Accordingly, the use of Shannon entropy should be understood as a measure of informational



structure within constrained Hilbert spaces, rather than as a direct statement about thermodynamic entropy in physical systems.

B Quantum Chemistry Hilbert Spaces as Structured Qubit Encodings

This appendix connects the constrained Hilbert-space structure of Sections 5–8 to concepts from quantum information theory.

The constrained Hilbert spaces described in the main text may also be interpreted from an information-processing perspective. In particular, the structure imposed by rotational symmetry, permutation symmetry, and non-Born–Oppenheimer correlation defines a restricted state space that can be viewed as an encoded subspace within a larger Hilbert space. This observation provides a natural point of contact with quantum information theory and, in particular, with the theory of fault-tolerant quantum computation.

B.1 Hilbert Space Embedding and Logical Subspaces

Let $\mathcal{H}_{\text{full}}$ denote the unconstrained tensor-product Hilbert space and $\mathcal{H}_{\text{phys}} \subset \mathcal{H}_{\text{full}}$ the symmetry-constrained physical subspace. The inclusion

$$\mathcal{H}_{\text{phys}} \hookrightarrow \mathcal{H}_{\text{full}} \quad (86)$$

may be interpreted as the definition of a logical subspace embedded within a larger ambient space.

In quantum information language, such a structure is analogous to an encoding in which logical states are restricted to a subspace defined by constraints. The constraints here are not imposed artificially, but arise from physical symmetries and many-body structure.

B.2 Symmetry Constraints as Stabilizing Conditions

Permutation symmetry and angular momentum conservation impose conditions that may be written schematically as

$$\hat{S}_k |\psi\rangle = \lambda_k |\psi\rangle, \quad (87)$$

where $\{\hat{S}_k\}$ represent symmetry operators (such as permutation operators or total angular momentum operators) and λ_k are the corresponding eigenvalues.

These relations define invariant subspaces:

$$\mathcal{H}_{\text{phys}} = \bigcap_k \ker(\hat{S}_k - \lambda_k I). \quad (88)$$

This structure is formally analogous to the stabilizer conditions used in quantum error correction,



where admissible states are defined as simultaneous eigenspaces of a set of commuting operators. The key distinction is that, in the present context, the operators arise from physical symmetry rather than engineered constraints.

B.3 Encoding Efficiency and State Reduction

The reduction in accessible state count,

$$N(\ell, N) \ll (2\ell + 1)^N, \quad (89)$$

implies that the effective logical space is smaller than the ambient space. In quantum information terms, this corresponds to an encoding in which a large physical space supports a smaller logical space.

The effective number of qubits associated with the constrained space may be defined as

$$n_{\text{eff}} = \log_2 N(\ell, N), \quad (90)$$

while the ambient space corresponds to

$$n_{\text{max}} = N \log_2(2\ell + 1). \quad (91)$$

The difference

$$n_{\text{max}} - n_{\text{eff}} \quad (92)$$

represents redundancy induced by symmetry constraints. Such redundancy is a central feature of fault-tolerant encoding schemes.

B.4 Error Structure and Constraint Preservation

Consider a perturbation that maps a physical state $|\psi\rangle \in \mathcal{H}_{\text{phys}}$ to a state in the larger space:

$$|\psi\rangle \rightarrow |\psi'\rangle = \hat{E} |\psi\rangle. \quad (93)$$

If $\hat{E}$ does not preserve the symmetry constraints, then $|\psi'\rangle \notin \mathcal{H}_{\text{phys}}$. In this sense, symmetry defines a notion of admissibility that distinguishes physical states from non-physical ones.

Operators that preserve symmetry satisfy

$$[\hat{E}, \hat{S}_k] = 0, \quad (94)$$

and therefore map admissible states to admissible states. Operators that violate symmetry act analogously to errors that move the system outside the logical subspace.

This distinction suggests a natural classification of perturbations into symmetry-preserving and



symmetry-violating classes, with the latter effectively detectable by projection onto $\mathcal{H}_{\text{phys}}$.

B.5 Relation to Fault-Tolerant Quantum Computation

Fault-tolerant quantum computation relies on encoding logical information into subspaces that are protected against certain classes of errors. The constrained Hilbert spaces arising in quantum chemistry share several structural features with such encodings:

1. **Subspace encoding:** physical states are restricted to $\mathcal{H}_{\text{phys}}$.
2. **Constraint operators:** symmetry operators define admissibility conditions.
3. **Redundancy:** the ambient space is larger than the logical space.
4. **Error detectability:** symmetry-violating perturbations move states outside the admissible subspace.

However, it is important to emphasize that the present framework does not constitute an explicit fault-tolerant scheme. Rather, it identifies a structural analogy between symmetry-constrained quantum systems and encoded subspaces in quantum information theory.

B.6 Interpretation

From this perspective, quantum chemical systems described in a fully non-Born–Oppenheimer framework may be viewed as naturally occurring constrained quantum systems in which admissible states form a structured subspace. The same symmetry principles that reduce entropy and induce fractal-like scaling also define admissibility conditions that resemble those used in encoded quantum computation.

This observation suggests that the study of symmetry-constrained Hilbert spaces may provide insight into structured quantum information representations, particularly in settings where constraints arise intrinsically rather than through external encoding procedures.

C Measurement and Projection in Constrained Hilbert Spaces

This appendix builds on the symmetry constraints introduced in Sections 4 and 6.

The measurement postulate of quantum mechanics states that the measurement of an observable A is associated with a spectral decomposition

$$A = \sum_i a_i P_i, \tag{95}$$



where $\{P_i\}$ are orthogonal projection operators onto the eigenspaces corresponding to eigenvalues $\{a_i\}$. For a system in state $|\psi\rangle$, the probability of obtaining outcome a_i is

$$p_i = \langle \psi | P_i | \psi \rangle, \quad (96)$$

and the post-measurement state is proportional to $P_i |\psi\rangle$.

In the framework developed in the main text, physical states are restricted to a constrained Hilbert space

$$\mathcal{H}_{\text{phys}} \subset \mathcal{H}_{\text{full}}, \quad (97)$$

defined by rotational symmetry, permutation symmetry, and full many-body correlation. This restriction modifies the effective action of measurement in a natural way.

C.1 Restricted Projection Operators

Although the spectral projectors $\{P_i\}$ are defined on the full Hilbert space, only their restriction to the physical subspace is relevant. Define

$$P_i^{\text{phys}} = P_i|_{\mathcal{H}_{\text{phys}}}. \quad (98)$$

If an eigenstate associated with P_i lies entirely outside $\mathcal{H}_{\text{phys}}$, then

$$P_i^{\text{phys}} = 0, \quad (99)$$

and the corresponding measurement outcome has zero probability for all physical states.

Thus, the admissible measurement outcomes are those for which the associated eigenspaces intersect nontrivially with $\mathcal{H}_{\text{phys}}$.

C.2 Constraint-Induced Selection of Outcomes

Let $\{|\phi_i\rangle\}$ denote a complete set of eigenstates of A in $\mathcal{H}_{\text{full}}$. The subset of physically admissible eigenstates is

$$\{|\phi_i\rangle \in \mathcal{H}_{\text{phys}}\}. \quad (100)$$

Measurement probabilities are therefore effectively supported only on this subset. For any $|\psi\rangle \in \mathcal{H}_{\text{phys}}$,

$$p_i = \langle \psi | P_i | \psi \rangle = 0 \quad (101)$$

whenever $|\phi_i\rangle \notin \mathcal{H}_{\text{phys}}$.

This establishes that symmetry constraints do not merely restrict state preparation, but also restrict the spectrum of physically realizable measurement outcomes.



C.3 Entropy of Measurement Outcomes

The Shannon entropy associated with measurement outcomes is

$$H_{\text{meas}} = - \sum_i p_i \log p_i. \quad (102)$$

Because the support of $\{p_i\}$ is restricted to admissible states, one has

$$H_{\text{meas}} \leq \log N(\ell, N), \quad (103)$$

with strict inequality in the presence of non-uniform probability distributions.

In contrast, the unconstrained Hilbert space would permit entropy up to

$$H_{\text{max}} = \log(2\ell + 1)^N. \quad (104)$$

Thus, symmetry constraints reduce not only the number of admissible states but also the entropy of observable measurement outcomes.

C.4 Relation to Fractal State Structure

The fractal-like structure discussed in the main text implies that admissible states occupy a recursively defined and asymptotically sparse subset of the ambient Hilbert space. Measurement therefore samples from this subset rather than from the full space.

In this sense, the measurement process may be viewed as probing a structured subset of Hilbert space whose effective dimension is smaller than that of the ambient space. The entropy of measurement outcomes reflects this reduced dimensionality, consistent with the scaling relations developed earlier.

C.5 Interpretation

The measurement postulate itself remains unchanged. However, when applied to systems with symmetry-constrained Hilbert spaces, its operational consequences are modified. Measurement outcomes are restricted to those compatible with symmetry, and the associated probability distributions reflect the reduced and structured nature of the admissible state space.

This perspective reinforces the interpretation of symmetry as a fundamental constraint not only on state construction and dynamics, but also on observable information extraction.



D Combinatorial Structure, Binomial Expansion, and Recursive State Growth

The recursive construction of many-body Hilbert spaces exhibits a combinatorial structure that is closely related to binomial expansion. This appendix clarifies that relationship and distinguishes between unconstrained combinatorial growth and the symmetry-constrained structure developed in the main text.

D.1 Binomial Expansion and Pascal Structure

The binomial expansion

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k \quad (105)$$

generates coefficients $\binom{n}{k}$ that are organized in Pascal's triangle. These coefficients encode the number of distinct ways to select k elements from a set of n , and arise naturally in recursive constructions where each step branches into multiple possibilities.

The defining recursion,

$$\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}, \quad (106)$$

expresses the fact that each level of the triangle is constructed from the previous one through additive combination of adjacent elements. This recursive rule provides a canonical example of combinatorial structure underlying recursive branching [28].

D.2 Tensor Product Growth in Hilbert Space

Consider a system of N degrees of freedom, each associated with an angular momentum sector of dimension $(2\ell + 1)$. The unconstrained Hilbert space is

$$\mathcal{H}_{\text{full}} = \bigotimes_{i=1}^N \mathcal{H}^{(i)}, \quad (107)$$

with dimension

$$\dim(\mathcal{H}_{\text{full}}) = (2\ell + 1)^N. \quad (108)$$

This structure is formally analogous to the expansion of a multinomial expression. Each additional degree of freedom corresponds to a branching step in which the number of possible configurations multiplies by a factor of $(2\ell + 1)$. In this sense, the growth of the unconstrained Hilbert space follows a combinatorial pattern closely related to binomial and multinomial expansion.



D.3 Angular Momentum Coupling and Multiplicity Structure

When angular momenta are coupled, the tensor product does not remain in factorized form. Instead,

$$\mathcal{H}_{\ell_1} \otimes \mathcal{H}_{\ell_2} = \bigoplus_{L=|\ell_1-\ell_2|}^{\ell_1+\ell_2} \mathcal{H}_L, \quad (109)$$

with multiplicities determined by the number of coupling pathways.

For multiple subsystems, the number of ways to obtain a given total angular momentum L grows combinatorially. These multiplicities arise from recursive coupling and exhibit a branching structure analogous to the recursive construction of Pascal's triangle, though generalized to the representation theory of $SO(3)$.

Thus, prior to the imposition of additional constraints, the structure of angular momentum coupling generates a combinatorial hierarchy in which multiplicity spaces play a role analogous to binomial coefficients.

D.4 Symmetry Constraints as Combinatorial Pruning

The introduction of permutation symmetry and other constraints modifies this combinatorial structure. The physically admissible Hilbert space satisfies

$$\mathcal{H}_{\text{phys}} \subset \mathcal{H}_{\text{full}}, \quad (110)$$

with

$$N(\ell, N) = \dim(\mathcal{H}_{\text{phys}}) \ll (2\ell + 1)^N. \quad (111)$$

From a combinatorial perspective, this reduction may be viewed as a pruning operation acting on the branching structure generated by tensor products and angular momentum coupling. While the unconstrained construction resembles Pascal-type recursive growth, symmetry constraints remove entire branches, leading to a reduced and structured subset of admissible states.

D.5 Relation to Fractal Scaling

The distinction between unconstrained combinatorial growth and symmetry-constrained state space is central to the emergence of fractal-like scaling. The unconstrained structure exhibits exponential growth, while the constrained structure satisfies

$$N(\ell, N) \sim (2\ell + 1)^{\alpha N}, \quad \alpha < 1. \quad (112)$$

This reduction in scaling exponent reflects the cumulative effect of recursive pruning. The resulting state space retains a hierarchical, recursively generated structure, but occupies a vanishing fraction of the ambient combinatorial space at large scale.



In this sense, Pascal-type combinatorial growth provides a useful reference model for the unconstrained case, while the symmetry-constrained Hilbert space developed in the main text represents a modified structure in which recursive branching is systematically restricted.

D.6 Interpretation

Pascal's triangle and binomial expansion illustrate the combinatorial structure that arises from repeated branching without constraint. The many-body Hilbert space, prior to symmetry projection, follows a closely related pattern. The imposition of symmetry transforms this structure by removing incompatible configurations, producing a reduced space with nontrivial scaling behavior.

The connection is therefore not one of direct geometric equivalence, but of underlying combinatorial structure: unconstrained quantum state construction exhibits binomial-type recursion, while the physically admissible space corresponds to a symmetry-filtered subset of that recursive expansion.

E Permutation Symmetry, Young Tableaux, and Representation Structure

This appendix refines the permutation symmetry analysis of Section 6 using the representation theory of S_N . In particular, Young tableaux provide a systematic combinatorial framework for organizing the multiplicity structure arising from the recursive angular momentum coupling described in Section 7.

The role of permutation symmetry in many-body quantum systems may be understood more precisely through the representation theory of the symmetric group S_N .

Young tableaux provide a systematic method for classifying irreducible representations of the symmetric group S_N [23] for organizing the combinatorial structure of tensor product spaces under symmetry constraints.

E.1 Irreducible Representations of S_N

For a system of N identical particles, the permutation group S_N acts on the Hilbert space by permuting particle labels. The irreducible representations of S_N are indexed by partitions of N , which may be represented graphically by Young diagrams.

Each Young diagram corresponds to a symmetry type, and the associated Young tableaux define basis states transforming under that irreducible representation. In particular:

- Fully symmetric states correspond to a single-row diagram.
- Fully antisymmetric states correspond to a single-column diagram.
- Mixed symmetry states correspond to more general partitions.



E.2 Young Symmetrizers and Projection Operators

Given a Young tableau T , one may construct an associated projection operator (Young symmetrizer) that projects onto the corresponding irreducible subspace:

$$\mathcal{Y}_T = a_T b_T, \quad (113)$$

where a_T symmetrizes over rows and b_T antisymmetrizes over columns.

Application of these operators to the tensor product space yields subspaces transforming irreducibly under S_N :

$$\mathcal{H}_{\text{phys}} = \bigoplus_{\lambda} \mathcal{H}_{\lambda}, \quad (114)$$

where λ labels partitions of N .

E.3 Connection to Quantum States

In quantum systems, physically admissible states must satisfy exchange symmetry requirements. For fermionic particles, only antisymmetric representations are allowed, while bosonic systems admit symmetric representations. More generally, internal degrees of freedom may lead to mixed symmetry types.

Young tableaux therefore provide a precise combinatorial framework for determining which components of the tensor product Hilbert space survive under symmetry constraints.

E.4 Relation to Combinatorial Growth

The unconstrained tensor product space exhibits combinatorial growth analogous to multinomial expansion. Young tableaux refine this structure by partitioning the space into symmetry sectors with well-defined multiplicities.

The dimension of each irreducible sector is given by the hook-length formula:

$$\dim(\lambda) = \frac{N!}{\prod_{(i,j) \in \lambda} h_{ij}}, \quad (115)$$

where h_{ij} are the hook lengths associated with the Young diagram λ [23].

This provides an explicit combinatorial count of symmetry-admissible configurations, replacing the naive binomial-type counting with a structured and constrained enumeration.

E.5 Symmetry-Resolved Hilbert Space Structure

The decomposition into irreducible representations implies that the Hilbert space is not only reduced in dimension, but also partitioned into sectors with distinct symmetry properties. These sectors



are not uniformly populated under physical constraints, and many are eliminated entirely by the requirements of indistinguishability and angular momentum coupling.

Thus, Young tableaux provide a refinement of the combinatorial structure discussed in Appendix D, replacing unconstrained Pascal-type recursion with symmetry-resolved branching.

E.6 Implications for Scaling

The restriction to specific irreducible representations contributes directly to the reduction in effective state count:

$$N(\ell, N) \ll (2\ell + 1)^N. \quad (116)$$

Because the admissible representations form only a subset of all possible partitions, and because additional constraints further restrict allowable states, the resulting Hilbert space exhibits reduced scaling behavior.

In combination with the recursive construction described earlier, this restriction contributes to the emergence of sublinear scaling and effective dimensional reduction.

E.7 Interpretation

Young tableaux provide the natural language for describing how permutation symmetry reorganizes and constrains the combinatorial growth of many-body Hilbert spaces. When combined with angular momentum coupling and other symmetry constraints, they define a structured subset of states whose organization reflects both combinatorial and representation-theoretic principles.

This framework clarifies that the reduction in accessible states is not arbitrary, but arises from well-defined symmetry structures acting on an underlying combinatorial expansion.

F Constraint, Entropy, and Internal Alignment: A Conceptual Perspective

This appendix extends the constrained-state framework of the main text to systems exhibiting internal state selection.

The formal developments presented in this work describe how symmetry constrains the structure of admissible Hilbert spaces, reducing both the number of accessible states and the effective entropy associated with those states. While these results arise from physical symmetries, they suggest a broader interpretive framework when considering systems capable of internal selection.

In general, a system embedded in a high-dimensional space of possibilities is not required to explore that space uniformly. Instead, it may exhibit structured behavior in which only a subset of available configurations is realized. In the language of information theory, this corresponds to a probability distribution that is concentrated on a restricted region of the admissible space, leading to reduced



effective entropy.

From this perspective, constraint is not merely a limitation, but a defining feature of structure. A system that imposes constraints on its own evolution is, in effect, selecting a trajectory through a space of possibilities rather than sampling that space arbitrarily. This distinction becomes particularly relevant when considering systems that exhibit adaptive, self-referential, or decision-making behavior.

One may interpret such behavior as a form of internal alignment: the system preferentially occupies configurations that are consistent with its internal organization. This alignment does not reduce the total number of possible states in the ambient space, but it reduces the set of states that are actually realized. The result is a form of stability in which the system maintains coherence even in the presence of external variability.

In environments characterized by increasing complexity or disorder, the number of available configurations grows. However, a system with sufficiently strong internal constraints need not reflect this increase in its realized behavior. Instead, it may continue to occupy a structured subset of states, effectively reducing the impact of external entropy on its internal configuration.

Within this framework, the concept of alignment may be understood as the maintenance of constraint under conditions of increasing combinatorial complexity. A system that preserves its internal constraints does not eliminate disorder in its environment, but it prevents that disorder from propagating into its realized state distribution.

This perspective suggests a reinterpretation of entropy in systems capable of internal selection. Rather than viewing entropy solely as a measure of disorder, one may also consider the relationship between entropy and constraint. High entropy in the ambient space corresponds to a large number of available configurations, while reduced entropy in the realized distribution reflects the presence of structure.

In this sense, coherence may be associated with the persistence of constraint, and instability with its breakdown. A system that loses its internal constraints becomes increasingly sensitive to external variability, while a system that maintains them preserves a stable trajectory through its state space.

Although the present work does not require any assumptions beyond standard quantum mechanics and information theory, this interpretation provides a conceptual bridge to systems in which internal structure governs behavior. In such systems, the reduction of effective entropy may be understood as a consequence of constraint-driven selection rather than a violation of the broader tendency toward increasing entropy in the environment.

This distinction is essential: entropy may increase globally, while remaining bounded locally within systems that maintain internal constraint. The interplay between these two regimes—global growth and local structure—may be viewed as a fundamental feature of systems capable of sustained organization.

The preceding discussion has treated entropy in terms of state counting and probability distributions over constrained Hilbert spaces. A complementary perspective arises when considering systems that do not explore all admissible states uniformly, but instead exhibit internally structured selection



among available configurations.

Let $\mathcal{H}_{\text{phys}}$ denote the admissible Hilbert space and $\{p_i\}$ a probability distribution over its states. The associated Shannon entropy is

$$H = - \sum_i p_i \log p_i. \quad (117)$$

In the absence of additional structure, the system may be approximated as sampling broadly across accessible states, leading to an entropy approaching.

However, if the system imposes internal constraints that restrict the set of admissible or preferred states, the effective entropy is reduced:

$$H_{\text{eff}} < \log N(\ell, N). \quad (118)$$

From an information-theoretic perspective, this corresponds to a reduction in uncertainty and an increase in specificity of the system's realized configurations.

F.1 Internal Constraint as Alignment

The presence of internal constraints may be interpreted as a form of alignment condition governing the system's evolution through its admissible state space. Rather than occupying states arbitrarily, the system preferentially selects configurations consistent with its internal structure.

In this sense, alignment may be understood not as an external imposition, but as an intrinsic constraint that reduces the effective dimensionality of the system's accessible configurations. The resulting behavior is characterized by a narrowing of the probability distribution and a corresponding reduction in entropy.

F.2 Stability Under External Disorder

In environments characterized by high combinatorial complexity, the number of available microstates increases, and the ambient entropy correspondingly grows. However, a system that maintains internal constraint structure need not reflect this increase in its realized configurations.

Instead, one may have

$$H_{\text{eff}} \ll H_{\text{max}}, \quad (119)$$

even when the surrounding space of possibilities expands.

This indicates that internal constraint structures can produce stability with respect to external disorder. The system does not eliminate entropy in the ambient space, but reduces its influence on the configurations that are actually realized.



F.3 Interpretation

Within this framework, the reduction of effective entropy arises from constraint rather than from suppression of underlying complexity. The system remains embedded in a high-dimensional space of possibilities, but occupies a structured subset defined by its internal conditions.

In systems capable of adaptive or self-referential behavior, this constraint may be interpreted as an alignment with a preferred subset of configurations. Such alignment does not alter the combinatorial structure of the full Hilbert space, but governs the trajectory of the system within it.

This perspective suggests that the relationship between entropy and structure is not solely determined by the size of the accessible space, but also by the presence of constraints that shape the distribution of realized states. In this sense, reduced entropy may reflect not a reduction in possibility, but an increase in coherence within the space of possibilities.

F.4 Extension to Systems with Internal Selection

The structure described above does not depend on the physical nature of the system alone, but on the presence of constraints that govern admissible configurations and their selection. While the preceding discussion has focused on quantum systems, the same formal structure arises in any system capable of restricting its trajectory through a high-dimensional space of possibilities.

In particular, systems that exhibit internal selection—where not all accessible configurations are equally realized—may be described in terms of effective entropy reduction within an otherwise large combinatorial space. The reduction arises not from elimination of possibilities in the ambient space, but from restriction of the realized distribution.

Such systems may maintain coherent behavior even under conditions of increasing external complexity. As the number of available configurations grows, the system does not necessarily reflect that growth in its realized states. Instead, it may continue to occupy a structured subset defined by internal constraints.

F.5 Interpretation in Terms of Human Experience

One class of systems exhibiting internal selection consists of those capable of adaptive and self-referential behavior. In such systems, decision processes may be viewed as trajectories through a space of possible configurations, where internal constraints influence which configurations are realized.

Within this perspective, coherence of behavior corresponds to persistence of constraint across changing conditions. The system does not eliminate external variability, but reduces its impact on the configurations that are actually selected.

This interpretation does not require any modification of the underlying physical framework. Rather, it reflects the possibility that the same mathematical structure—recursive construction, constrained admissibility, and entropy reduction through selection—can describe both physical state spaces and



systems that exhibit internally guided evolution.

F.6 Conceptual Bridge

The connection established here is therefore structural rather than ontological. The constrained Hilbert spaces developed in the main text provide a precise mathematical model of how admissible configurations are restricted by symmetry and interaction. Systems capable of internal selection exhibit an analogous restriction, in which the realized subset of configurations is governed by internal constraints rather than external symmetry alone.

In both cases, the result is a reduction in effective entropy relative to the ambient space, accompanied by persistence of structure under conditions of increasing combinatorial complexity. This parallel suggests that the relationship between constraint, entropy, and structure may extend beyond strictly physical systems to include systems characterized by internally guided state selection.

These considerations provide a structural perspective on proposals in which cognition or consciousness is associated with constrained state selection in high-dimensional spaces. In such interpretations, coherence may be understood not as unrestricted superposition, but as the persistence of trajectories within symmetry-compatible subsets of the admissible Hilbert space.

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